Mock Interview: Kafka

Quiz ?

Statement

How does Kafka scale its brokers, topics, producers, and consumers?

Answer

Topics: Kafka scales topics by spreading them over many partitions. Each partition can be placed on a separate broker.
Producers: By having multiple partitions of a topic, Kafka producers can publish a huge number of messages to that topic. Multiple producers can publish messages on the same topic.
Consumers: Kafka scales consumers by partition such that each consumer gets its share of partitions. A consumer can have more than one partition, but a partition can only be used by one consumer in a consumer group at a time.
Broker: Adding or removing brokers is very easy. Whenever a new broker is added or removed, Kafka rebalances topic partitions. Kafka tries to distribute leaders and followers of a partition on separate brokers.

Hide Answer

Quiz ?

Statement

How does Kafka ensure fault-tolerance and reliability of controller broker, topic partition leader, and ZooKeeper?

Answer

Broker & Partition: Kafka is designed in such a way that a broker failure is detectable by ZooKeeper and other brokers in the cluster. Because each topic can have multiple followers running on different brokers, the cluster can recover from any broker failures and continue to work without any disruption of service.
Controller & Leader: Whenever a controller broker or a partition leader stops, ZooKeeper detects the failure and initiates a new leader election.
ZooKeeper: Since all the critical information is stored in the ZooKeeper, and it replicates this data across its cluster, ZooKeeper's failure does not affect the state of the Kafka cluster. Kafka will restore the state, once the ZooKeeper restarts after failure.

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Quiz ?

Statement

What message-delivery guarantees does Kafka provide to producers?

Answer

Kafka provides three types of message delivery guarantees to producers:

1. **Async**: Producer sends a message to Kafka and does not wait for an acknowledgment from the server. 2. **Committed to Leader**: Producer waits for an acknowledgment from the partition leader. 3. **Committed to Leader and Quorum**: Producer waits for an acknowledgment from the partition leader and the full set of in-sync replicas.

Hide Answer

Quiz ?

Statement

What message-delivery guarantees does Kafka provide to consumers?

Answer

Kafka provides three types of message delivery guarantees to consumers:
At-most-once: Messages may be lost but are never redelivered. **At-least-once**: Messages are never lost but may be redelivered. **Exactly-once**: Each message is delivered once and only once.

Hide Answer

Quiz ?

Statement

How does Kafka ensure consistency as far as message delivery is concerned?

Answer

To ensure data consistency, leader brokers never return messages which have not been replicated to a minimum set of in-sync replicas (ISR). Brokers keep track of the **high-water mark offset**, which is the largest offset that a minimum set of ISRs of a particular partition share.

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Quiz ?

Statement

How does Kafka ensure durability?

Answer

Kafka ensures durability by replicating messages across multiple brokers, as well as storing them on the disk.

Hide Answer

Quiz ?

Statement

How Kafka manages high throughput for consumers?

Answer

By using consumer groups, consumers can be parallelized, so that multiple consumers can read from multiple partitions on a topic, allowing a very high message processing throughput.

Hide Answer

Quiz ?

Statement

How does Kafka manage high availability?

Answer

Each topic partition can have multiple replicas (called followers). If the leader of a partition stops, a follower can take over to make the partition available.
If a broker dies, Kafka distributes its partitions to other available brokers.
If the controller broker dies, ZooKeeper initiates a leader election to choose a new controller broker.

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Quiz ?

Statement

How does Kafka throttle its producers and consumers?

Answer

Kafka throttles its clients using **quotas**, which are byte-rate thresholds defined per client-ID. A client-ID logically identifies an application making a request. A single client-ID can span multiple producer and consumer instances. The quota is applied for all instances as a single entity. When the broker calculates that a client has exceeded its quota, it slows the client down by holding back the response to the client for enough time to keep the client under the quota.

Hide Answer

Quiz ?

Statement

How does Kafka perform failover for consumers within a consumer group?

Answer

If a consumer in a consumer group dies, the partitions assigned to that consumer are distributed to the remaining consumers in that group.

Hide Answer

Quiz ?

Statement

How does Kafka perform failover for brokers?

Answer

If a broker dies, Kafka divides up the leadership of its topic partitions to the remaining brokers in the cluster.

Hide Answer

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